

YadYar Lite

T-05

Retrieval-Augmented Generation and the Study of Hallucination

Final Project Report - Phases 1 and 2

Course Artificial Intelligence, Spring 1404-1405

Dataset SQuAD 2.0

Baseline all-MiniLM-L6-v2 + FAISS + google/flan-t5-small

Evaluation 30 questions from a 300-context closed corpus

Prepared July 2026

A lightweight, reproducible comparison of RAG and No-RAG behavior, with retrieval evaluation, generation metrics, and representative hallucination analysis.

Executive Summary

This project asks whether a lightweight Retrieval-Augmented Generation (RAG) pipeline can reduce unsupported answers relative to the same generator used without retrieval. The baseline is intentionally small and explainable: SQuAD 2.0 supplies question-context pairs, all-MiniLM-L6-v2 creates dense embeddings, FAISS ranks 300 indexed contexts, and google/flan-t5-small generates answers. No fine-tuning, training from scratch, heavy statistical testing, or user-interface development is performed.

Phase 2 evaluates the unchanged Phase 1 baseline on 30 questions: 23 answerable and 7 unanswerable. The gold context is present in the index by construction, so retrieval results measure ranking quality inside a controlled closed corpus rather than open-domain retrieval performance.

Main result

RAG clearly outperformed No-RAG on lexical answer quality (EM 0.333 vs. 0.000; Token F1 0.441 vs. 0.000). However, neither mode abstained on any of the seven unanswerable questions. Retrieval was strong (Recall@1 0.913), but correct evidence did not guarantee a correct answer: ignored-evidence errors remained the dominant answerable failure mode.

Finding	Observed result	Interpretation
Top-1 retrieval	21 of 23 answerable questions	The dense retriever usually ranked the gold context first.
RAG answer quality	EM 0.333; Token F1 0.441	Retrieved evidence substantially improved the small generator.
No-RAG answer quality	All reported lexical scores were 0	Without evidence, FLAN-T5-small produced unrelated short answers.
Unanswerable behavior	0 of 7 questions correctly abstained	The baseline is not safe against absent-evidence hallucination.
Dominant error	10 ignored-evidence cases	Generation remains a bottleneck even when retrieval succeeds.

1. Project Question and Scope

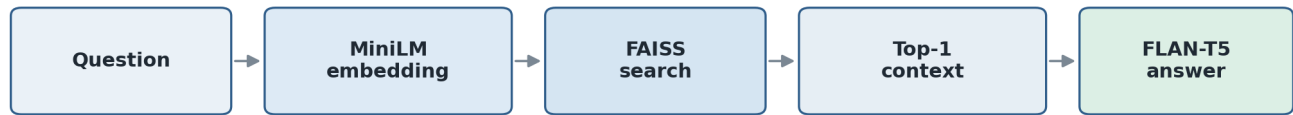
Research question: Can a lightweight dense-retrieval RAG pipeline reduce unsupported or hallucinated answers compared with a no-retrieval baseline on a small SQuAD 2.0 subset?

The comparison is controlled: both modes use the same google/flan-t5-small generator and greedy decoding. The only substantive difference is whether the generator receives the retrieved context. This makes the experiment easy to reproduce and suitable for the project requirement of a focused, undergraduate-level ML study.

- In scope: pretrained models, a small public dataset, simple retrieval and generation metrics, representative errors, a notebook demo, and a future-integration contract.
- Out of scope: model training, full fine-tuning, a new architecture, formal hypothesis testing, production deployment, or a polished user interface.

2. System Design and Phase 1 Baseline

RAG path: retrieval supplies evidence to the same generator used by the No-RAG baseline



No-RAG: question -> FLAN-T5

Figure 1. Baseline architecture. The No-RAG mode skips retrieval and sends only the question to the same generator.

Component	Choice	Reason for the lightweight baseline
Dataset	rajpurkar/squad_v2, train split	Contains answerable and unanswerable questions with gold contexts.
Embedding	sentence-transformers/all-MiniLM-L6-v2	Small, fast, 384-dimensional sentence embeddings.
Vector search	FAISS IndexFlatL2	Exact and explainable search over only 300 context vectors.
Generator	google/flan-t5-small	Instruction-tuned, CPU-friendly, and unchanged across RAG/No-RAG.
Decoding	Greedy (no sampling)	Deterministic outputs support reproducibility.

3. Dataset and Evaluation Subset

Phase 1 builds a deterministic corpus of 300 unique contexts from the shuffled SQuAD 2.0 train split (seed 42). Questions are considered unanswerable when `answers["text"]` is empty; the code does not depend on an `is_impossible` column. Phase 2 selects 30 questions whose gold context exists in the Phase 1 index, preserving the original baseline and avoiding synthetic data.

Subset property	Value
Indexed contexts	300
Evaluated questions	30
Answerable	23 (76.7%)
Unanswerable	7 (23.3%)
RAG evidence supplied	Top-1 retrieved context
Retrieval diagnostics	Top-1, top-3, top-5 and reciprocal rank

Important validity boundary

Because each evaluated gold context is guaranteed to be inside the 300-context index, Recall@k measures ranking within a selected closed corpus. It must not be presented as retrieval performance over the full SQuAD dataset or an open document collection.

4. Evaluation Protocol

Metric	Definition and role
Recall@k	Fraction of answerable questions whose exact gold context

Metric	Definition and role
	appears in the first k retrieved contexts.
MRR	Mean of 1/rank for the gold context; rewards placing relevant evidence near the top.
Exact Match	Normalized exact string match against any gold answer; unanswerable rows require abstention.
Token F1	Best token-overlap F1 across all gold answers; gives partial credit.
ROUGE-L	Longest-common-subsequence overlap; another lexical partial-credit metric.
Unanswerable detection	Fraction of unanswerable questions where the model produces a recognized abstention phrase.

5. Retrieval Results

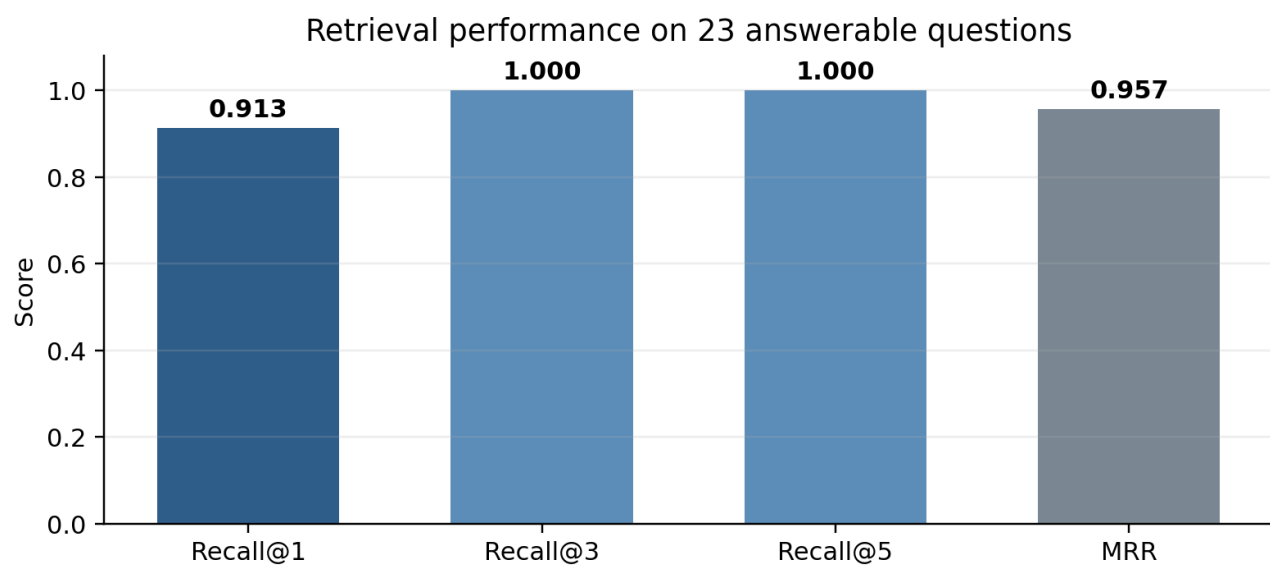


Figure 2. Retrieval metrics on the 23 answerable questions.

Metric	Value	Meaning
Recall@1	0.913	The gold context was ranked first for 21 of 23 answerable questions.
Recall@3	1.000	Both rank-1 misses were recovered by the first three results.
Recall@5	1.000	All gold contexts were present in the first five results.
MRR	0.957	Relevant contexts were usually first and otherwise still near the top.

The retrieval module is strong for this controlled corpus. The metric most directly aligned with generation is Recall@1 because Phase 1 passes only the top-ranked context to the generator. Recall@3 and Recall@5 are diagnostic: they show that a lightweight reranker or multi-context prompt could potentially recover the two top-1 failures, but those alternatives were not added because Phase 2 preserves the original baseline.

6. Generation Results: RAG vs. No-RAG

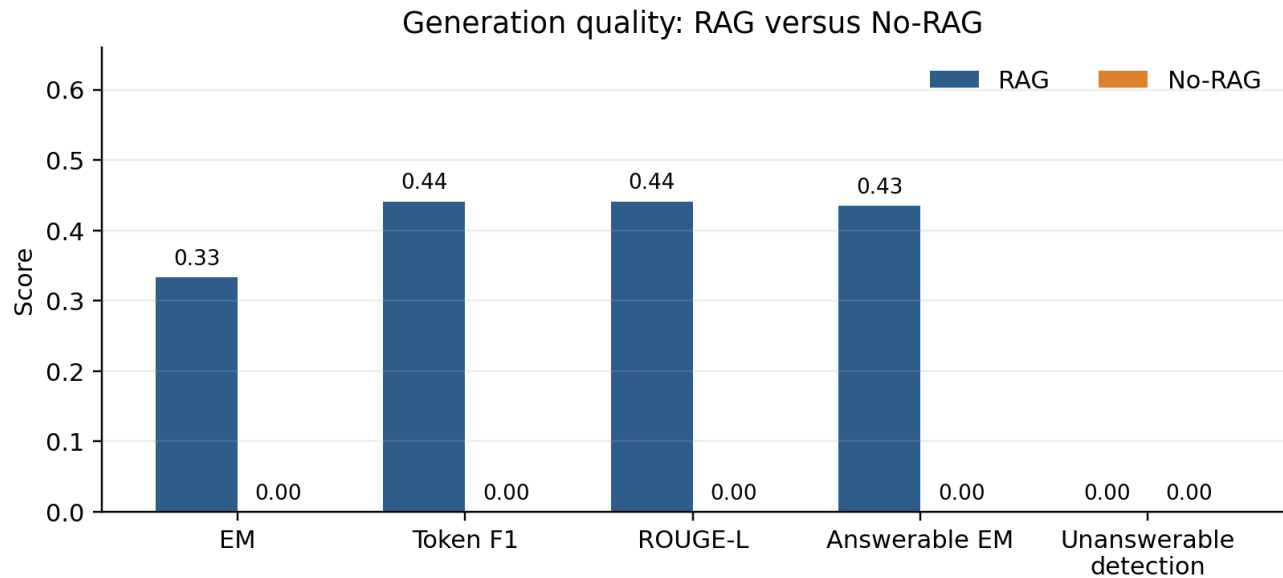


Figure 3. Lexical generation metrics for the unchanged RAG and No-RAG modes.

Mode	EM	Token F1	ROUGE-L	Answerable EM	Unanswerable detection
RAG	0.333	0.441	0.441	0.435	0.000
No-RAG	0.000	0.000	0.000	0.000	0.000

RAG provides a clear benefit: ten of the thirty predictions exactly match a gold answer after normalization, while No-RAG receives no lexical credit. The all-zero No-RAG result is unusual but consistent with the saved raw predictions, which include unrelated answers such as “a few days” for a gold answer of “250 years.” It should therefore be reported as an observed baseline outcome rather than smoothed or replaced.

7. Simple Result Breakdown

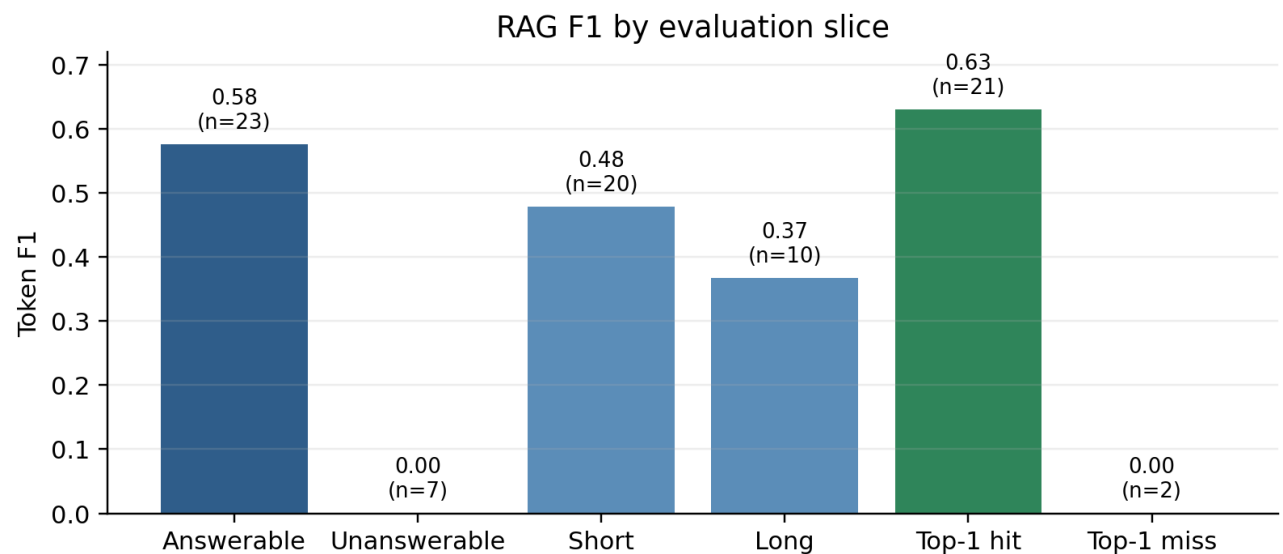


Figure 4. RAG Token F1 across the required simple evaluation slices.

Slice	n	RAG EM	RAG F1	No-RAG EM	No-RAG F1
Answerable	23	0.435	0.575	0.000	0.000
Unanswerable	7	0.000	0.000	0.000	0.000
Short (≤ 10 words)	20	0.350	0.478	0.000	0.000

Slice	n	RAG EM	RAG F1	No-RAG EM	No-RAG F1
Long (> 10 words)	10	0.300	0.367	0.000	0.000
Answerable & retrieval success (top-1)	21	0.476	0.630	0.000	0.000
Answerable & retrieval failure (top-1)	2	0.000	0.000	0.000	0.000

The strongest evidence of the retrieval-generation dependency is the last pair of rows: answerable questions with a top-1 retrieval hit reached F1 0.630, while the two top-1 misses received zero. Question length had a smaller effect than retrieval success. All unanswerable scores were zero because neither model abstained.

8. Hallucination and Representative Error Analysis

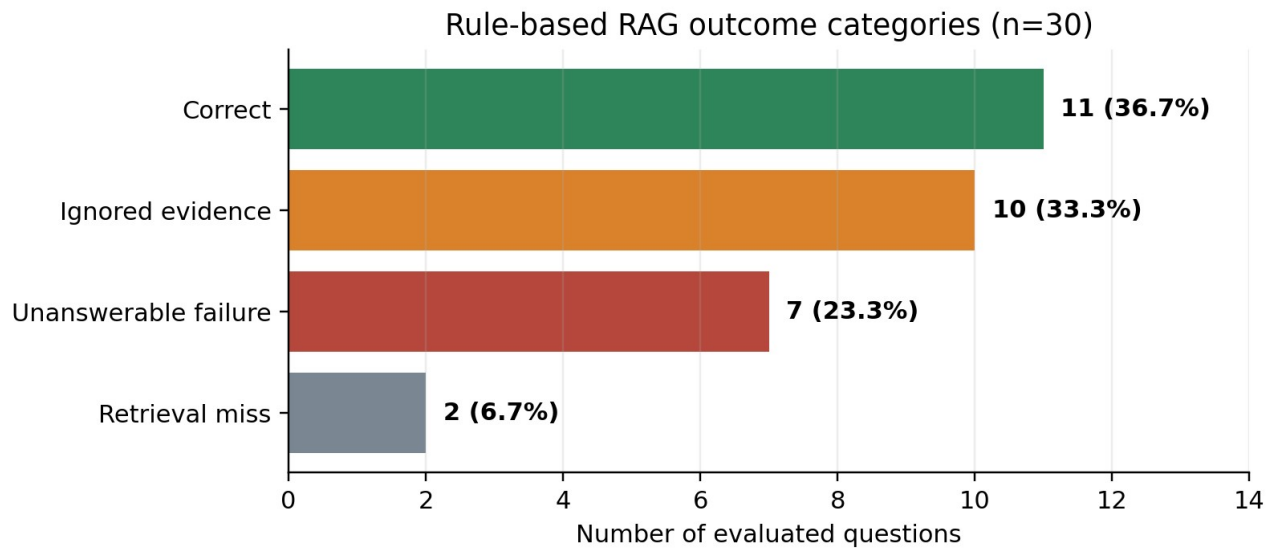


Figure 5. Rule-based categories assigned to all 30 RAG outputs.

The transparent classifier labels an output as correct when it exactly matches or has high token overlap with a gold answer. A wrong answer is then attributed to a top-1 retrieval miss, ignored evidence when the gold context was retrieved, or an unanswerable failure. These labels are useful for inspection but are not equivalent to expert human annotation.

Question (shortened)	Gold	RAG answer	Category	Why it happened
How long did it take for Myanmar to recover from the collapse of its first kingdom?	250 years	250 years	Correct	Directly extracted from the correct top-1 context.
What was the republican government amenable to?	war reparations	assassinations of political activists	Retrieval miss	Gold context ranked second; the generator answered from unrelated top-1 evidence.
How many casualties did the US Air Force suffer during WWII?	68,000	3	Ignored evidence	The gold context was top-1, but the generator selected the wrong number.
What does OCA stand for?	Offensive Counterair	Counterair	Partial / ignored evidence	The answer is related but incomplete; lexical F1 gives partial credit.
In what year did Heinz Kloss develop a framework for describing dialects genetically?	Unanswerable	1967	Unanswerable failure	The context mentions 1967 but does not support the full wording of the question.
Which recitation is the basis of the Cairo Quran?	'Asim's	Asim's recitation	Metric artifact / near-correct	Unicode apostrophe and an added word cause lexical scoring to under-credit a semantically close answer.

9. Interpretation of the Main Failure Modes

- Ignored-evidence hallucination is the largest answerable failure group (10 of 30 total rows). Retrieval alone does not guarantee faithful generation.
- All seven unanswerable questions produced content answers. The prompt's abstention instruction is too weak for this small generator.
- Only two answerable questions suffered a top-1 retrieval miss, but both received zero EM/F1; better use of ranks 2-3 could help.
- Lexical metrics can be harsh on near-correct forms, abbreviations, Unicode punctuation, and answers that add a short explanatory phrase.

10. Discussion

The experiment supports the narrow project claim: adding retrieved evidence substantially improves this generator relative to asking the same model to answer from its parametric knowledge. The difference is large enough to be visible even in a 30-question run. At the same time, the results do not show that the system is dependable. A 91.3% top-1 retrieval rate coexists with only 43.5% answerable EM, demonstrating that generation quality is the main bottleneck after retrieval succeeds.

The No-RAG scores should be interpreted cautiously rather than dismissed. FLAN-T5-small is a very small instruction-tuned model, the questions are fact-specific, and the no-context prompt provides no evidence. The raw outputs show coherent but unrelated short completions; zero EM/F1 is therefore plausible. A stronger generator could be an optional comparison, but it must not replace the required Phase 1 baseline.

11. Limitations

- Small evaluation size: 30 questions is adequate for a lightweight course study but too small for strong statistical claims.
- Closed-corpus selection: the gold context is always indexed, making retrieval metrics optimistic relative to a realistic open corpus.
- Weak generator: FLAN-T5-small exposes failure modes clearly but limits absolute answer quality and abstention behavior.
- Single fixed prompt and top-1 context: no prompt tuning, reranking, multi-context generation, or answerability classifier was used.
- Automatic evaluation is lexical: EM, F1, and ROUGE-L do not fully measure semantic correctness or evidence support.
- Hallucination labels and abstention detection are rule-based and can misclassify near-correct or unusually phrased answers.
- CPU-oriented execution and library/model versions may cause small runtime differences, although greedy decoding improves reproducibility.

12. Future Work

Priority	Proposed improvement	Observed motivation
1	Strengthen abstention prompting or add a separate answerability gate	Both modes failed all seven unanswerable questions.
2	Add a lightweight reranker or supply the top 2-3 contexts	Recall@3 was 1.0, so rank-2/3 evidence could recover top-1 misses.
3	Try a stronger embedder such as BGE-base or E5-base	Test whether semantic retrieval remains strong on a larger corpus.
4	Evaluate a stronger generator as an optional comparison	Separate limitations of RAG from limitations of FLAN-T5-small.
5	Run limited human review and Unicode-aware normalization	Improve faithfulness judgments and reduce lexical metric artifacts.

13. Lightweight Demo and Future Integration

The notebook demo prints three real SQuAD examples: two answerable and one unanswerable. For each question it displays retrieved contexts and similarity scores, the RAG answer, the No-RAG answer, and the gold answer. No separate UI is required.

Contract	Example
Input	<code>{"question": "...", "course_documents": ["document text", "..."]}</code>
Output	<code>{"answer": "...", "retrieved_contexts": ["..."], "confidence_or_warning": "Evidence may be insufficient", "error_type_if_known": "..."}}</code>

In YadYar Lite, the component could sit downstream of document parsing and upstream of dialogue or quiz generation. The output should retain retrieved evidence and a warning field so downstream components can avoid presenting unsupported answers as facts.

14. Reproducibility and Delivered Artifacts

- Run the combined notebook from the first Phase 1 cell through the final Phase 2 cell in Google Colab.
- Core packages: datasets, sentence-transformers, faiss-cpu, transformers, pandas, and standard Python utilities.
- Saved outputs: phase2_results.csv, representative_errors.csv, metrics_summary.json, retrieval_metrics.json, breakdown_results.csv, the Phase 2 Markdown report, README, and ZIP package.
- Reproducibility controls: fixed seed, pinned Hugging Face model IDs, exact FAISS index, and greedy generation.

15. Conclusion

The project meets its lightweight objective and answers its research question in a qualified way. RAG markedly improves the same small generator over No-RAG, but the system still fails when evidence is absent and often mishandles evidence that was retrieved correctly. The most defensible conclusion is therefore not that retrieval solves hallucination, but that retrieval is necessary and insufficient: the next practical improvement should target abstention and faithful evidence use rather than heavy model training.

Final takeaway

RAG changed the baseline from zero lexical accuracy to meaningful answer quality, while the 0% unanswerable detection rate exposed the system's central safety limitation. This combination provides a clear, defensible Phase 2 finding.

External AI Tools Acknowledgement

External AI assistants were used for drafting, debugging, code review, and writing support. All code, outputs, metric values, and interpretations were reviewed by the team. No synthetic data or invented experimental results were used. The team remains responsible for understanding and defending every submitted component.

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